One might suggest another solution to this game. Each player knows that he cannot count on his opponent to play the strategy yielding the equilibrium value he, the first player, prefers; thus each should assume that the other player will not play that strategy and then pick his best strategy under *that* assumption. In our example this means that Able will go skiing and Baker will go to the festival.

Accepting this proposal, however, forces us to conclude that the pair (R_1, C_2) with the value (-1, -1) is the solution to the game, which is scarcely the best possible outcome. Can we endorse canons of rationality that drive individuals to outcomes that are clearly not optimal? Before examining this question let us consider another game in which the issue arises even more dramatically.

PROBLEMS

1. Show that the clash of wills can be fully represented without using utility numbers.
2. Explain why, if the two players coordinated their weekend by flipping a fair coin, it would make sense to assign the value 3/2 to each of them.
3. Show that if Able plays ($1/2\,R_1$, $1/2\,R_2$), Baker can do better than 3/2 by playing C_2.
4. In the rural community where I live, falling trees and lightning cause a fair number of electrical power failures. When these occur someone phones the electric company to report the outage and it is fixed within a few hours. Every time I have called the electric company I have been told that someone has already reported the outage and it will be repaired. Since the call is a long-distance one, I have felt that I wasted my money. The other night we had another power failure, and as I lay on my bed staring into the darkness, it occurred to me that in deciding whether to phone the company, I was playing a version of the clash of wills. Assuming that only one other person has been affected by the power failure, and thus that he is my opponent, represent my situation as a clash of wills.

5-4b. The Prisoner's Dilemma

The next game has a single equilibrium point and dominant strategies for each player. Thus, unlike the clash of wills, its outcome is fully predictable. The paradox is that although from the point of view of the individual players their chosen strategies are the only rational alternatives, the resulting outcome is patently nonoptimal. This suggests that there may be a fundamental conflict between the notions of individual and group rationality.

The game is named after a story about two prisoners. They have been arrested for vandalism and have been isolated from each other. There is sufficient evidence to convict them on the charge for which they have been arrested, but the prosecutor is after bigger game. He thinks that they robbed a bank together and that he can get them to confess to it. He summons each separately to an interrogation room and speaks to each as follows: "I am going to offer the same deal to your partner, and I will give you each an hour to think it over before I call you back. This is it: If one of you confesses to the bank robbery and the other

does not, I will see to it that the confessor gets a one-year term and that the other guy gets a twenty-five year term. If you both confess, then it's ten years apiece. If neither of you confesses, then I can only get two years apiece on the vandalism charge; but that's not going to happen—at least one of you will confess." Sure enough, an hour later both confess, and both end up serving ten years. Let us see why they were forced to that outcome.

The game can be represented by table 5-31. Here I have used the length of the prisoners' sentences to assign numerical outcomes. It is easy to see that

5-31	Confess	Do Not
Confess	(−10, −10)	(−1, −25)
Do Not	(−25, −1)	(−2, −2)

confessing dominates for both players, and thus the (−10, −10) outcome will be the one that our theory predicts "rational" players will achieve. But clearly *both* would be better off if they could achieve the outcome (−2, −2). But (−10, −10) and *not* (−2, −2) is the equilibrium value, and the strategies producing it are recommended by one of our first and most fundamental principles of individual rationality—the dominance principle. The paradox is that following the dictates of individual rationality has put everyone in a worse position than they would have been if they had been less "rational."

Let us look at the prisoner's dilemma from an informal point of view. The prisoner, either one, knows that if neither confesses, both will be better off than if both confess. So he might consider not confessing. But he also knows that if he does not confess, he has placed himself in an extremely vulnerable position; a "double-crossing" partner could cost him twenty-five years of his life. Each prisoner suspects that his partner is thinking in the same way about the matter, and each sees that each has a strong reason to try for the double-cross. Both conclude that the only thing to do is to play it safe, but playing it safe means confessing.

As the informal version of the dilemma shows, no numerical reasoning is necessary to see how the game will go. Thus, we require only purely ordinal utility functions to represent this game, and need not concern ourselves with interpersonal comparisons of utility.

5-4c. Other Prisoner's Dilemmas

The situation arising in the prisoner's dilemma is not restricted to two-person games. Suppose the members of a cartel, OPEC, for instance, try to limit the total sales of their product in the hope, of course, of driving up prices and benefiting individually and as a group. Once prices start to rise, each member of the cartel will be sorely tempted to accept contracts at the higher prices and violate the production quotas assigned by the cartel. For, so long as he is the only one to do this, prices will not drop and nobody will try to repudiate their contracts with him. On the other hand, if enough members of the cartel try the same double-crossing strategy, there will be a glut on the market, contracts will

be broken, and prices will tumble. Since this is known to all cartel members, they will find it nearly impossible to limit production in the first place. This multiperson prisoner's dilemma in which multiple double-crosses are possible is not far from real life.

One might be tempted to argue that criminals will *not* confess because each knows that double-crossers are killed by their fellow criminals. No doubt this is a factor in many real-life confrontations between criminals and prosecutors. But when this happens we do not have a true prisoner's dilemma game. For the outcome of confessing when one's partner does not is not just a one-year sentence; it is one year plus a good chance of being killed. Presumably, this is less preferable than a simple ten-year sentence. Thus the dominance argument no longer works.

Similarly, the members of a cartel know that they must deal with each other again and again. Some mutual trust is of the essence; thus there is a premium on not double-crossing. This can prevent the game from being a true prisoner's dilemma.

Ironically, one of these games based on an element of trust can develop into a prisoner's dilemma if the players know that mutual trust is no longer important to them as individuals. Suppose, for example, that over the years two competing stores located on the same block have, through mutual trust, avoided all-out price wars. Now suppose the entire block is to be demolished for an urban renewal project, and they will move to different locations never to compete again. Then trust is no longer a factor in their dealings. Each has a strong reason to cut prices just enough to draw away the competitor's customers. We have a prisoner's dilemma. It would be no wonder, then, if both began liquidation sales.

Moreover, if both competitors know the exact date on which they must close their doors, then no matter how cooperative they have been until now, they can reason themselves backward to a prisoner's dilemma right now. For suppose all occupants of the block must vacate in forty-five days. Then each knows that on the forty-fifth day it can no longer trust its competitor to maintain its prices. Trust being out of the question on day forty-five, each figures it might as well cut its prices on day forty-four to get the jump on the competitor. But since each knows that, trust is out of the question on day forty-four. Thus each reckons it must cut its prices by day forty-three. But then trust on day forty-three is out of the question; so we are back to day forty-two. This reasoning continues until each store concludes that it should cut prices today!

Such reasoning can even force competing stores to lower their prices to the point where both operate at a near loss. Suppose that two stores, X and Y, currently sell a given commodity for \$2 and make a \$.50 profit. Store X decides to cut its prices to \$1.90 to attract some of Y's customers, figuring with the added volume to exceed its previous profits based on a lower volume and a higher per sale profit. Store X responds by cutting its prices to \$1.80; Y responds by going to \$1.70, and if X then sets its prices at \$1.60, they will both end up making the same number of sales but at only one fifth the profit.

PROBLEMS

1. Show that the prisoner's dilemma arises when Col's sentences are as before